Question 1: (a) $[18] = \{ 4n + 2 \mid n \in \mathbb{Z} \}$.

(b) $[31] = \{ 4n + 2 \mid n \in \mathbb{Z} \}$.

(c) There are 4 equivalence classes.

Question 2: *Proof.* We prove that $\equiv_n$ on $\mathbb{Z}$ is an equivalence relation by showing that it is reflexive, transitive, and symmetric. Suppose $n \in \mathbb{Z}^+$.

Reflexive Let a be an arbitrary element in $\mathbb{Z}$. Clearly, $a - a = 0 = kn, k \in \mathbb{Z}$. So, $\forall a \in \mathbb{Z}, a \equiv_n a$.

Symmetric Let a, b be arbitrary elements of $\mathbb{Z}$. Suppose $a \equiv_n b$. By definition, $a - b = kn, k \in \mathbb{Z}$. Multiplying the equation by -1, we get

$$b - a = -kn.$$

$-k$ is still an integer, and thus by definition, $b \equiv_n a$.

Transitive Let a, b, c be aribtrary elements of $\mathbb{Z}$. Suppose $a \equiv_n b$ and $b \equiv_n c$. By definition, we have $a - b = kn, k \in \mathbb{Z}$ and $b - c = gn, g \in \mathbb{Z}$. Then, $b = a - nk$. Subbing in, we get that

$$\begin{gathered} a - kn - c = gn \\ a - c = (k + g)n \end{gathered}$$

By definition, we have $a \equiv_n c$.

□

Question 3: *Proof.* Suppose $a, b, c, d \in \mathbb{Z}$, and $a \equiv_n c$ and $b \equiv_n d$. Then,

$$\begin{gathered} a - c = nk, k \in \mathbb{Z} \\ b - d = nl, l \in \mathbb{Z}. \end{gathered}$$

Adding the equations, we get

$$a + b - c - d = n(k + l).$$

Simplifying, we get

$$(a + b) - (c + d) = n(k + l).$$

Thus, by definition $a + b \equiv_n c + d$. □

Question 4: (a) *Proof.* The theorem given in the proposition is equivalent to claiming that 'If ab is even, then a is even or b is even'. We have proven this theorem to be true, so it is the case that if $ab \equiv_2 0$, then $a \equiv_2 0$ or $b \equiv_2 0$. □

(b) The answer is no. $10 \times 2 = 20 \equiv_4 0$, but $10 \not\equiv_4 0$ and $2 \not\equiv_4 0$.

(c) The answer to the question will be yes when n is a prime.

Proof. Suppose n is prime. Let $a, b \in \mathbb{Z}$. We proceed via the contrapositive, which is the statement

$$\text{if } a \not\equiv_n 0 \text{ and } b \not\equiv_n 0, \text{ then } ab \not\equiv_n 0.$$

Then, n is not a prime factor of either a or b. □

Question 5: *Proof.* We show three requirements.

1 $e \in A$ is its own inverse ($e * e = e$), and so it has an inverse, and is in $A^\times$.

2 Suppose $a \in A^\times$. Then, $a \in A$ and a has an inverse $a^{-1} \in A$. Similarly, a^{-1} has an inverse, namely a. So, $a^{-1} \in A$ and a^{-1} is invertible, so $a^{-1} \in A^\times$.

3 Suppose $a, b \in A^\times$. Then, $a, b \in A$, and $a * b \in A$. Let $c = (b^{-1} * a^{-1})$. Then, we compute

$$(a * b) * c = a * b * b^{-1} * a^{-1} = a * a^{-1} = e.$$

Therefore, $a * b$ has an inverse. Therefore, $a * b \in A^\times$.

□

Question 6: *Proof.* We first show that matrix multiplcation $(*)$is a unital binary operation on $\mathrm{Mat}_{2\times 2}(\mathbb{R})$.

1 We are given that $*$ is associative.

2 Let $E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{Mat}_{2\times 2}(\mathbb{R})$. We compute that

$$AE = EA = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Thus, $*$ is unital.

Next, we show that the set $\mathrm{GL}_2(\mathbb{R})$ is actually the set of all the matrices in $\mathrm{Mat}_{2\times 2}(\mathbb{R})$ that are invertible. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. We then reduce A to E augmented with E:

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right]$$

Let the resultant augmented matrix A'. Then, we compute

$$AA' = E \text{ and } A'A = E.$$

A' exists only when $ad - bc \neq 0$, which is true for all $m \in \mathrm{GL}_2(\mathbb{R})$. In other words, $\mathrm{GL}_2(\mathbb{R}) = \{\, A \in \mathrm{Mat}_{2\times 2}(\mathbb{R}) \mid A \text{ is invertible} \,\}$.

Lemma 1. *See the result of problem 5.*

Thus, $\mathrm{GL}_2(\mathbb{R})$ is a group with respect to matrix multiplication. □

Question 7:

(a) We show three requirements.

Reflexive $*$ is unital, so there is an identity element e. $e * e = e$, so e is invertible and is its own inverse. So, $a = e * a * e$. Therefore, R is reflexive.

Symmetric Let $a, b \in A$ and aRb. Then, we compute

$$\begin{aligned} b &= t^{-1} * a * t \\ t^{-1} * b &= a * t \\ t^{-1} * b * t &= a. \end{aligned}$$

Let $t^{-1} = t'$. $t' \in A$ and is invertible. Thus, bRa, and R is symmetric.